

BHAGALPUR, IndiaWMO: **424980**Lat: **25.233N**Lon: **86.950E**Elev: **49**StdP: **100.74**Time Zone: **5.50 (IND)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	11.8	8.3	6.8	13.7	9.2	7.3	14.6	2.5	17.9	2.3	17.7	0.4	N/A	0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.6	38.7	27.3	37.6	27.2	36.5	27.3	30.3	35.4	29.7	34.3	29.3	33.7	0.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	26.0	33.2	28.6	25.2	32.4	28.2	24.6	32.0	102.6	35.3	99.0	34.8	97.3	34.3	33.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.4	2.2	1.8	8.4	40.4	1.2	2.1	7.5	41.9	6.8	43.1	6.2	44.3	5.3	45.8
			8.4	31.0	1.5	1.4	7.3	32.0	6.5	32.8	5.7	33.6	4.6	34.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	16.2	20.7	26.4	30.1	31.1	31.6	30.0	30.3	29.9	27.7	23.3	17.9
	DBStd	5.62	2.30	2.49	2.41	2.29	2.33	2.13	1.61	1.59	1.74	2.05	2.18	2.43
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	114	69	6	0	0	0	0	0	0	0	0	0	38
	CDD10.0	5946	193	301	508	603	654	648	621	629	597	549	398	243
	CDD18.3	3019	4	74	250	353	396	398	362	371	347	291	148	24
	CDH23.3	38449	8	363	2631	5027	5815	5929	4883	5074	4645	3179	840	53
	CDH26.7	19689	0	68	1097	2862	3432	3551	2419	2602	2306	1189	159	5
Wind	WSAvg	0.4	0.3	0.4	0.5	0.7	0.7	0.5	0.5	0.4	0.3	0.2	0.2	0.2
Precipitation	PrecAvg	1164	12	12	12	23	67	151	309	264	219	70	9	5
	PrecMax	1904	41	38	60	75	231	331	493	448	616	308	101	51
	PrecMin	725	0	1	0	0	0	8	159	109	56	2	0	0
	PrecStd	272	12	11	14	26	58	94	98	101	143	75	21	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.4	30.5	36.2	39.0	40.5	40.0	36.7	36.6	35.9	35.2	31.5	27.4
		MCWB	18.9	22.2	23.8	25.3	26.4	28.5	29.6	29.9	29.8	28.2	24.1	22.1
	2%	DB	22.5	28.7	34.7	37.6	39.0	38.4	35.0	35.3	34.7	33.4	29.9	24.8
		MCWB	17.8	20.9	23.6	24.4	26.5	29.1	29.0	29.4	29.1	27.4	23.6	19.5
	5%	DB	21.3	27.0	33.1	36.7	37.6	37.4	33.9	34.1	33.8	32.2	28.4	23.1
		MCWB	17.6	20.1	23.1	24.9	27.0	29.0	28.7	28.9	28.7	27.0	23.4	19.2
	10%	DB	20.2	25.4	31.7	35.5	36.2	36.1	32.8	33.1	32.9	31.2	27.2	21.8
		MCWB	17.1	19.4	22.9	25.3	27.3	28.8	28.4	28.6	28.3	26.7	22.9	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	23.6	27.3	30.2	31.1	32.2	30.1	30.2	30.2	29.2	25.4	22.6
		MCDB	22.7	29.0	32.7	36.2	36.6	37.6	35.1	35.0	34.8	33.4	29.9	26.2
	2%	WB	18.9	22.1	25.7	28.7	29.6	30.7	29.5	29.6	29.5	28.3	24.5	20.5
		MCDB	21.2	26.6	31.0	34.9	35.3	35.8	33.6	33.7	33.6	32.0	28.5	23.3
	5%	WB	18.1	21.1	24.8	27.8	28.9	29.7	29.1	29.2	29.1	27.7	23.9	19.4
		MCDB	20.6	25.5	30.0	33.9	34.6	34.6	32.7	32.9	32.8	31.2	27.7	22.3
	10%	WB	17.4	20.2	24.0	27.0	28.3	29.3	28.7	28.9	28.6	27.2	23.2	18.6
		MCDB	19.8	24.4	29.6	32.6	33.7	34.3	31.9	32.3	32.1	30.4	26.6	21.4
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.5	9.1	9.0	8.0	6.6	4.7	4.5	4.8	5.8	6.5	6.1
		MCDBR	9.2	10.9	12.1	10.6	10.6	9.1	6.5	6.5	6.6	7.8	8.7	8.6
		MCWBR	5.6	5.6	5.6	4.9	4.3	3.6	2.5	2.4	2.7	3.3	3.9	4.8
	5% WB	MCDBR	7.5	9.7	10.2	9.6	9.3	7.8	5.6	5.7	5.9	6.3	7.2	7.6
		MCWBR	5.1	5.4	5.3	4.9	4.4	3.6	2.6	2.5	2.8	3.1	3.5	4.5
Clear-Sky Solar Irradiance	taub	0.591	0.548	0.557	0.626	0.699	0.699	0.601	0.570	0.580	0.593	0.651	0.651	
	taud	1.678	1.811	1.825	1.739	1.648	1.669	1.892	1.991	1.942	1.846	1.648	1.619	
	Ebn at Noon	647	732	755	714	662	658	725	747	727	686	594	575	
	Edh at Noon	218	203	210	235	256	249	199	180	185	194	223	224	
All-Sky Solar Radiation	RadAvg	3.31	4.50	5.63	6.20	6.00	5.16	4.41	4.60	4.49	4.44	3.92	3.21	
	RadStd	0.30	0.28	0.22	0.31	0.38	0.36	0.32	0.27	0.26	0.36	0.21	0.32	